Wall time per step: Splat simulation

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

100k (96,342 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells	CyteArc mtx	CyteArc h5ad
load	9.7	5.5	5.1	4.6	18	2.3	5.1	34	4.8	47	37	8.5	13	32	11
qc	0.1	0.1	0.1	0.1	0.6	0.6	0.9	0.4	0.4	1.3	2.6	2.8	1.1	0.0	0.0
normalise	n/a	n/a	n/a	n/a	2.0	1.9	0.0	0.1	0.1	0.0	0.2	0.2	0.4	2.9	2.9
hvg	0.4	0.3	0.3	0.2	1.4	1.4	1.4	0.1	0.1	3.8	3.7	3.8	2.9	2.2	2.3
pca	0.8	0.8	0.9	0.5	2.2	2.2	3.9	2.0	2.0	13	10	11	58	6.0	6.1
harmony	0.5	0.5	0.9	0.3	0.5	0.5	0.5	12	14	12	13	13	14	14	13
knn_snn	2.7	2.7	1.9	1.5	27	27	24	6.5	6.3	6.8	7.5	7.6	7.9	24	24
cluster	0.9	1.0	0.8	0.8	3.0	2.7	2.7	15	13	13	20	20	21	14	14
cluster_fast	1.0	1.0	0.4	0.3	n/a	n/a	n/a	18	17	17	n/a	n/a	n/a	n/a	n/a
umap	3.8	3.6	0.9	0.8	39	38	38	37	38	38	46	47	47	84	84
dge	0.6	0.6	0.7	0.5	47	46	–	3.3	3.3	81	11	11	57	6.3	6.2
tsne_default	43	43	37	34	162	167	155	n/a	n/a	n/a	126	125	142	n/a	n/a
tsne_fft	15	15	3.2	3.1	38	38	36	n/a	n/a	n/a	33	33	32	n/a	n/a
tsne_quick	11	11	4.6	3.4	n/a	n/a	n/a	20	21	21	n/a	n/a	n/a	n/a	n/a
total (no tSNE)	19	15	12	9.2	141	122	77	112	83	216	151	125	223	186	164

500k (482,409 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells	CyteArc mtx	CyteArc h5ad
load	48	29	21	19	OOM	11	21	203	18	228	216	35	65	156	37
qc	0.3	0.2	0.3	0.3	–	2.8	3.8	2.2	2.2	6.3	13	13	4.0	0.1	0.1
normalise	n/a	n/a	n/a	n/a	–	3.6	0.0	0.4	0.4	0.0	0.8	0.8	1.9	13	13
hvg	1.0	1.1	0.8	0.7	–	3.1	4.5	0.5	0.5	19	19	19	13	7.2	7.2
pca	2.3	2.3	1.3	1.1	–	10	11	15	15	71	OOM	OOM	274	23	28
harmony	1.7	1.6	1.1	0.7	–	2.1	2.0	100	83	84	–	–	88	72	81
knn_snn	14	14	7.5	6.7	–	33	29	40	40	42	–	–	45	145	159
cluster	4.1	4.0	3.2	3.1	–	20	20	115	123	103	–	–	133	16	16
cluster_fast	14	14	3.1	2.5	n/a	n/a	n/a	196	194	197	n/a	n/a	n/a	n/a	n/a
umap	35	34	3.9	3.7	–	226	226	182	175	178	–	–	257	421	423
dge	2.3	2.3	1.8	1.9	–	454	–	22	23	1,717	–	–	260	13	13
tsne_default	252	251	223	225	–	1,315	1,319	n/a	n/a	n/a	–	–	1,390	n/a	n/a
tsne_fft	63	61	14	20	–	153	156	n/a	n/a	n/a	–	–	156	n/a	n/a
tsne_quick	42	48	27	23	n/a	n/a	n/a	132	133	135	n/a	n/a	n/a	n/a	n/a
total (no tSNE)	109	88	41	37	failed	766	317	681	481	2,448	failed	failed	1,140	866	777

1M (964,763 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells	CyteArc mtx	CyteArc h5ad
load	90	57	49	42	OOM	OOM	39	OOM	OOM	405	OOM	OOM	129	303	67
qc	0.5	0.4	0.4	0.4	–	–	7.2	–	–	12	–	–	8.6	0.1	0.1
normalise	n/a	n/a	n/a	n/a	–	–	0.0	–	–	0.0	–	–	4.1	27	28
hvg	2.0	2.2	1.6	1.5	–	–	8.2	–	–	38	–	–	25	13	14
pca	4.8	5.2	2.6	2.1	–	–	18	–	–	144	–	–	463	50	54
harmony	3.2	3.0	2.0	1.2	–	–	4.2	–	–	166	–	–	175	163	148
knn_snn	29	30	15	13	–	–	40	–	–	87	–	–	95	364	393
cluster	8.2	8.2	6.7	6.5	–	–	47	–	–	269	–	–	343	19	20
cluster_fast	46	47	7.9	6.3	n/a	n/a	n/a	–	–	545	n/a	n/a	n/a	n/a	n/a
umap	80	80	11	9.7	–	–	555	–	–	362	–	–	546	863	869
dge	4.1	4.2	3.6	3.3	–	–	–	–	–	TIMEOUT	–	–	518	24	24
tsne_default	654	628	562	544	–	–	TIMEOUT	n/a	n/a	n/a	–	–	TIMEOUT	n/a	n/a
tsne_fft	194	197	30	46	–	–	–	n/a	n/a	n/a	–	–	–	n/a	n/a
tsne_quick	134	134	84	73	n/a	n/a	n/a	–	–	–	n/a	n/a	n/a	n/a	n/a
total (no tSNE)	222	190	91	80	failed	failed	719	failed	failed	failed	failed	failed	2,305	1,826	1,617

per step: fastest … slowest

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.